ONCO-TWIN | System Architecture

Digital twin fusing static EHR and wearable time series to predict severe chemotherapy toxicity and choose the safest dosing window

Team Aurora · The Apollo University
Happiest Health Digital Twin Challenge 2026

1 DATA SOURCES
synthetic only

Synthetic Data Generator — seed 42
- Hidden 5-compartment Friberg marrow model per patient (nadir ~day 10-14)
- Label from latent ANC, NOT from wearables (no circular validation)
- Wearables = noisy, lagged proxies of marrow strain; Gompertz tumor + imaging noise

EHR / Patient Profiles — STATIC
- 150 patients: age, sex, breast cancer stage II-IV
- Biomarkers, baseline ECOG, baseline ANC

Clinical Logs — HISTORICAL
- 13,500 daily rows (90 days/pt): ANC, WBC
- 600 dosing events (AC / paclitaxel)
- Tumor volume measurements

Wearable Streams — DYNAMIC
- 13,500 daily rows (90 days/pt)
- Resting HR, HRV (RMSSD), step count, sleep duration (Apple Health / Fit style)

2 FEATURE PIPELINE
src/features.py

Zero Future Lookahead Gate — LEAK-FREE
- Every lab and wearable input frozen at day D (only data ≤ D)
- Candidate delay enters only as a known scheduling variable
- Verified by perturbation invariance test (atol 1e-7)

Clinical Features (Ablation A, 8)
- age, baseline_ecog, baseline_anc, pre_dose_anc
- tumor_volume_cm3
- delay_days, days_since_last_dose, dose_interval_bin_8_14

Wearable Features (+8 = B, 16)
- Today: resting_heart_rate, hrv_rmssd, step_count, sleep_duration_mins
- Trend: 7-day vs 30-day personal-baseline ratios for RHR, HRV, steps, sleep

Output Matrices
- processed_dataset.csv (1,500 rows, training)
- timing_features.csv (1,200 rows, +0/+3/+7/+14d)

3 DIGITAL TWIN STATE
virtual patient at day D

Virtual Patient (≤ D)

Profile
demographics, stage, biomarkers, ECOG

Treatment state
last dose, days since dose, cycle

Lab state
latest ANC, WBC vs baseline

Tumor state
current volume, fitted growth and kill kinetics

Autonomic state
RHR, HRV, steps, sleep vs 30-day personal baseline

Re-scored every morning
as new wearables and labs arrive

4 PREDICTIVE ENGINES
src/train_toxicity.py · tumor_model.py

MAIN: Toxicity Risk Model — ADVERSE EVENT
- Standardized Logistic Regression (C=1.0), chosen over Random Forest and Gradient Boosting benchmarks
- Patient-grouped splits, repeated 5x5 GroupKFold, 5 data seeds
- Safety-tuned threshold ≈ 0.108 inside validation folds (not 0.50)
- OUTPUT: P(grade ≥ 3 neutropenia, ANC < 1.0, in the 21 days after dose or until next dose) for each candidate day +0 / +3 / +7 / +14, with top feature drivers

SUPPORT: Cost-of-Delay — TRADE-OFF
- Patient-specific growth (α) and chemo kill (κ) fitted by bounded nonlinear least squares on observations ≤ D
- $dV/dt = \alpha \cdot V \cdot \ln(K/V) - \kappa \cdot Dose \cdot V$
- OUTPUT: tumor volume forecast per delay (0/3/7/14 d) with 90% uncertainty band, so 'safest day' is not always 'later'

EXPERIMENTAL: Marrow twin — RESEARCH
- 5-compartment Friberg ODE + Kalman filter; hidden by default; delay trend only 49.8% consistent

5 CLINICIAN DASHBOARD
app/dashboard.py (Streamlit)

Patient + Day Selector
150 synthetic patients, any decision day D

Tab 1: Dosing Window
- Dual-axis chart: toxicity risk % vs tumor regrowth %
- 90% tumor forecast ribbon
- Action matrix: green < 0.11, amber 0.11-0.22, red > 0.22

Tab 2: Physiology
- ANC trajectory with grade 3/4 line
- HRV: baseline vs 7-day window

Tab 3: Model Performance
- Ablation table, ROC, calibration

ONCOLOGIST
Reviews drivers, explains rationale to patient, schedules lab verification, makes the final decision

DAILY RE-SCORING LOOP
New wearable and lab readings → twin state refreshed → risk and tumor forecast recomputed each morning, so the plan adapts as marrow recovery actually unfolds (retrospective features only).

VALIDATION AND QUALITY (synthetic cohort, multi-seed mean over 5 seeds)

0.891	+0.010	90.9% / 42.7%	0.32	4.74% vs 7.73%	98.1%	30 tests
holdout AUROC (5-seed mean)	wearable gain (-0.009 to +0.021)	recall / precision at tuned threshold	delay-ranking tau (weak, advisory only)	tumor MAPE: patient-fit vs population	coverage of 90% band	pytest incl. leakage checks; ruff clean

For clinical decision support and simulation only. Not an autonomous prescription engine. All data is synthetic; not clinically validated.

Stack: Python · Pandas · NumPy · SciPy · scikit-learn · Plotly · Streamlit · Pytest · Ruff | MIT License